**MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY**
**COLLEGE OF SCIENCE AND TECHNICAL EDUCATION**

**DEPARTMENT OF MATHEMATICS AND STATISTICS**
**SEMESTER I ACADEMIC YEAR 2025/2026**
**MS 8221: DIFFERENTIAL EQUATIONS**
**UQF 8 THIRD YEAR: COMPUTER ENGINEERING**
**LECTURE 1**
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# INTRODUCTION

**CONCEPTS:**

1) This module has two main parts which are:
   (a) Differential Equations
   (b) Complex Variables
2) The part of Differential Equations which covers the following topics:
   (a) Power Series Solution of Differential Equations
      (i) Legendre's Equation
      (ii) Frobenius Method
      (iii) Bessel's Equation
   (b) Partial Differential Equations
      (i) One Dimensional Wave Equation
      (ii) Heat Conduction Equation
      (iii) Laplace's Equation
      (iv) Solving Partial Differential Equations using Fourier Transforms
3) The part of Complex Variables covers the following topics:
   (a) Functions of a Complex Variable
   (b) Complex Mapping and Transformation
   (c) Differentiation of Complex Functions
      (i) Cauchy-Riemann Condition

(ii) Laurent Series

(d) Contour Integrals

(i) Bromwich Integral

(ii) Trigonometric Integrals of Real variable

4) We shall begin with the part of Differential Equations, but let us review on **Power Series**.

5) Now, here we go:

# PART A: DIFFERENTIAL EQUATIONS

## 1.0: Revision of Power Series

**CONCEPTS:**

1) Power Sries are the infinite series of the form:

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots + a_{n-1}x^{n-1} + a_n x^n + a_{n+1}x^{n+1} + \cdots \tag{1}$$

$$\sum_{n=0}^{\infty} a_n (x-x_0)^n = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + \cdots + a_{n-1}(x-x_0)^{n-1} + a_n(x-x_0)^n + \cdots \tag{2}$$

$\Rightarrow$ Where $a_0, a_1, a_2, \cdots, a_{n-1}, a_n, a_{n+1}, \cdots$ are constants called the **Coefficients** of the series, and $x_0$ is a constant called the **Center** of the series.

2) The first is called a power series in $x$, while the second is called a power series in $(x-x_0)$ or power series centered at $x_0$ or power series about $x_0$.

3) If a power series converges for some for some values of $x$, then the sum of the series is a function $f(x)$ given by

$$f(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots + a_n x^n + \cdots$$

Or

$$f(x) = \sum_{n=0}^{\infty} a_n (x-x_0)^n = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + \cdots + a_n(x-x_0)^n + \cdots$$

4) Notice that $f(x)$ resembles a **Polynomial** with the only difference that $f(x)$ has **infinitely** many terms.

## 1.1: Differentiation of Power Series/Derivatives of Power Series

**CONCEPTS:**

1) The sum of a power series is a function $f(x)$, so we differentiate such a function by differentiating each individual term in the series just as we do for polynomial.
$\Rightarrow$ This is called **Term by Term Differentiation**.

2) Thus if:

$$f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots + a_n x^n + \cdots = \sum_{n=0}^{\infty} a_n x^n$$

Or

$$f(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \cdots + a_n(x - x_0)^n + \cdots = \sum_{n=0}^{\infty} a_n(x - x_0)^n$$

$\Rightarrow$ Then we have;

(i) $f'(x) = a_1 + 2a_2 x + 3a_3 x^2 + \cdots + na_n x^{n-1} + (n+1)a_{n+1}x^n + \cdots$

$$\Rightarrow \quad f'(x) = \sum_{n=1}^{\infty} na_n x^{n-1}$$

(ii) $f''(x) = 2a_2 + 6a_3 x + 12a_4 x^2 + \cdots + (n+1)na_{n+1}x^{n-1} + (n+2)(n+1)a_{n+2}x^n + \cdots$

$$\Rightarrow \quad f''(x) = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2}$$

Or

(iii) $f'(x) = a_1 + 2a_2(x - x_0) + 3a_3(x - x_0)^2 + \cdots + na_n x^{n-1} + (n+1)a_{n+1}(x - x_0)^n + \cdots$

$$\Rightarrow \quad f'(x) = \sum_{n=1}^{\infty} na_n (x - x_0)^{n-1}$$

(iv) $f''(x) = 2a_2 + 6a_3(x - x_0) + \cdots + (n+1)na_{n+1}x^{n-1} + (n+2)(n+1)a_{n+2}(x - x_0)^n + \cdots$

$$\Rightarrow \quad f''(x) = \sum_{n=2}^{\infty} n(n-1)a_n (x - x_0)^{n-2}$$

## 1.2: Taylor and Maclaurin Series

**CONCEPTS:**

1) Suppose that $f(x)$ is any function that can be represented by a power series:

$$f(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + a_3(x - x_0)^3 + \cdots = \sum_{n=0}^{\infty} a_n(x - x_0)^n \tag{1}$$

2) Now we need to determine what the coefficients $a_n$ must be in terms of $f(x)$.

3) To begin, notice that if we put $x = x_0$ in (1), then all terms after the first one are zero, and we get:

$$f(x_0) = a_0$$

4) Differentiating (1) term by term, we get:

$$f'(x) = a_1 + 2a_2(x - x_0) + 3a_3(x - x_0)^2 + 4a_4(x - x_0)^3 + \cdots \tag{2}$$

and then put $x = x_0$ in (2) gives:

$$f'(x_0) = a_1$$

5) Now differentiating (2), we get:

$$f''(x) = 2a_2 + (2)(3)a_3(x - x_0) + (3)(4)a_4(x - x_0)^2 + \cdots \tag{3}$$

again, we put $x = x_0$ in (3), we get:

$$f''(x_0) = 2a_2$$

6) Differentiating (3) gives:

$$f'''(x) = (2)(3)a_3 + (2)(3)(4)a_4(x - x_0) + (3)(4)(5)a_5(x - x_0)^2 + \cdots \tag{4}$$

and put $x = x_0$ in (4), we get:

$$f'''(x_0) = (2)(3)a_3$$

7) By now you can see the pattern. If we continue to differentiate and put $x = x_0$, we obtain:

$$f^{(n)}(x_0) = (2)(3)(4)\cdots(n)a_n = n!a_n$$
$$\Rightarrow\ f^{(n)}(x_0) = n!a_n$$

$\Rightarrow$ Solving this equation for the n$^{\text{th}}$ coefficient $a_n$, we get:

$$a_n = \frac{f^{(n)}(x_0)}{n!}$$

8) This formula remains valid even for $n = 0$, since $0! = 1$ and $f^{(0)} = f$. Thus, we have the following Theorem.

9) **THEOREM:**
$\Rightarrow$ If $f(x)$ has a power series expansion at $x = x_0$, that is, if

$$f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n$$

then its coefficients $a_n$ are given by:

$$a_n = \frac{f^{(n)}(x_0)}{n!}$$

10) Put $a_n$ into the series, then we see that if $f(x)$ has a power series expansion at $x_0$, it must be of the form:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

$$\Rightarrow \; f(x) = f(x_0) + \frac{f'(x_0)}{1!}(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \frac{f'''(x_0)}{3!}(x - x_0)^3 + \cdots \tag{5}$$

11) The series in (5) is called the **Taylor Series** of $f(x)$ at $x = x_0$ or about $x_0$ or centered at $x_0$.

12) For the special case $x_0 = 0$, the Taylor series becomes:

$$\Rightarrow \; f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots \tag{6}$$

13) The series in (6) arises most frequently and is called the **Maclaurin Series**.

## EXAMPLES

1. Find the Maclaurin series of the function $f(x) = e^x$.

## SOLUTION

$\Rightarrow$ If $f(x) = e^x$ then $f^{(n)}(x) = e^x$, so $f^{(n)}(0) = e^0 = 1$ for all $n$.

$\Rightarrow$ Therefore the Taylor series for $f(x)$ at 0 (that is, the Maclaurin Series) is:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$\Rightarrow$ **NOTE:** From $f(x) = e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$, we have;

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$\Rightarrow$ In particular, if we put $x = 1$, we get the expression for the number $e$ as a sum of an infinite series:

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots$$

2. Find the Taylor series for $f(x) = e^x$ at $x_0 = 2$.

**SOLUTION**

$\Rightarrow$ For $f(x) = e^x$ at $x_0 = 2$, we have, $f^{(n)}(2) = e^2$, thus putting $x_0 = 2$ in $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$, we get;

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(2)}{n!}(x-2)^n = \sum_{n=0}^{\infty} \frac{e^2}{n!}(x-2)^n$$

$\Rightarrow$ Therefore $f(x) = e^x = \sum_{n=0}^{\infty} \frac{e^2}{n!}(x-2)^n$ at $x_0 = 2$.

3. Find the Maclaurin series of the function $f(x) = \sin x$.

**SOLUTION**

$\Rightarrow$ We arrange our computations in two columns as follows:

$$\begin{aligned}
f(x) &= \sin x & f(0) &= 0 \\
f'(x) &= \cos x & f'(0) &= 1 \\
f''(x) &= -\sin x & f''(0) &= 0 \\
f'''(x) &= -\cos x & f'''(0) &= -1 \\
f^{(4)}(x) &= \sin x & f^{(4)}(0) &= 0 \\
f^{(5)}(x) &= \cos x & f^{(5)}(0) &= 1 \\
f^{(6)}(x) &= -\sin x & f^{(6)}(0) &= 0 \\
f^{(7)}(x) &= -\cos x & f^{(7)}(0) &= -1 \\
f^{(8)}(x) &= \sin x & f^{(8)}(0) &= 0
\end{aligned}$$

$\Rightarrow$ Since the derivatives repeat in a circle of four, we can write the Maclaurin series as follows:

$$f(x) = \sin x = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$

$$\Rightarrow \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + - \cdots = \sum_{n=0}^{\infty}(-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

4. Find the Maclaurin series of the function $f(x) = \cos x$.

**SOLUTION**

$\Rightarrow$ We could proceed directly as in Example 3, but it is easier to differentiate the Maclaurin series for $\sin x$. That is;

$$\begin{aligned}
\cos x = \frac{\mathrm{d}}{\mathrm{d}x}(\sin x) &= \frac{\mathrm{d}}{\mathrm{d}x}\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + - \cdots\right) \\
&= 1 - \frac{3x^2}{3!} + \frac{5x^4}{5!} - \frac{7x^6}{7!} + - \cdots
\end{aligned}$$

$$\Rightarrow \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + - \cdots = \sum_{n=0}^{\infty}(-1)^n \frac{x^{2n}}{(2n)!}$$

# 2.0: Power Series Solution of Differential Equations

**CONCEPTS:**

1) Linear second order variable coefficient differential equations arise in many applications, but only in few special cases it is possible to express their general solution of a finite linear combination of elementary functions.

2) That is, linear ordinary differential equations with constant coefficients can be solved by functions known from calculus such as sine, cosine, exponential and logarithm, all of whose mathematical properties are well known.

3) However, if a linear ordinary differential equation has variable coefficients (functions of $x$), it must usually be solved by other methods as we shall see in this topic.

4) The approach developed in this topic involves seeking solutions of linear second order variable coefficient differential equations in the form of **Power Series**, and in the other cases using an approach due to **Frobenius** that involves seeking solutions in the form of power series multiplied by a factor $x^c$, where $c$ is not an integer.

5) Applications are made to a number of typical linear variable coefficient differential equations, and then to the important **Legendre** and **Bessel** equations that lead in turn to **Legendre Polynomial** and **Bessel Functions**.

# 2.1: Power Series Methods of Differential Equations

**CONCEPTS:**

1) Power series method is the standard method for solving linear ordinary differental equations with variable coefficients, where it gives solutions in the form of power series.

2) From calculus, we recall that a power series in powers of $(x - x_0)$ is an infinite of the form:

$$\sum_{n=0}^{\infty} a_n(x-x_0)^n = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + \cdots \qquad (1)$$

3) In particular, if $x_0 = 0$, we obtain a power series of $x$ as:

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots \qquad (2)$$

4) Some of the familiar examples of power series are the Maclaurin Series:

   (i) $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$

   (ii) $\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - + \cdots$

   (iii) $\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - + \cdots$

5) The idear of power series method for solving ODEs is simple and natural, where we describe the practical procedure in the following three different appraches.

## 2.1.1: A First Approach to Power Series Solution of Differential Equations

**CONCEPTS:**

1) We begin our approach by showing how the series solution can be found for first and second order linear differential equations specified at $x = x_0$.
2) The series we obtain will be in powers of $(x - x_0)$, and they will be said to be expanded about the point $x_0$.
3) The first order linear differential equation will be assumed to be of the form:

$$y' + p(x)y = r(x) \qquad \text{with} \quad y(x_0) = y_0 \tag{1}$$

and the second order linear differential equation will be of the form:

$$y'' + P(x)y' + Q(x)y = R(x) \qquad \text{with} \quad y(x_0) = y_0, \quad y'(x_0) = y_1 \tag{2}$$

$\Rightarrow$ Where, the functions $p(x), r(x), P(x), Q(x)$ and $R(x)$ can all be expanded as **Taylor Series** about the point $x_0$.

4) Functions which have a Taylor series expansion about $x_0$ are said to be **Analytic** at $x_0$. That is, a function $f(x)$ is analytic at $x_0$ if:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

5) Polynomials, $\sin x, \cos x$ and $e^x$ are **analytic** everywhere, so are sums, differences and products of these functions. Quotients of any two of these are analytic at all points where the denominator is not zero.
6) The point $x_0$ is an **Ordinary** point of the differential equations (1) and (2) if all $p(x), r(x), P(x), Q(x)$ and $R(x)$ are analytic at $x_0$.

$\Rightarrow$ If either of these functions is not analytic at $x_0$, then $x_0$ is called a **Singular Point** of the differential equations (1) and (2).

7) The method to be developed will be seen to be capable of extension to a higher order linear differential equations provided only that the coefficients of $y$ and its derivatives that are involved and the nonhomogeneous term are analytic at $x_0$.
8) Now the approach is best illustrated by considering the differential equation (1), and seeking a solution about $x_0$ of the form:

$$y(x) = y(x_0) + y'(x_0)(x - x_0) + \frac{y''(x_0)}{2!}(x - x_0)^2 + \frac{y'''(x_0)}{3!}(x - x_0)^3 + \cdots \tag{3}$$

9) Setting $x = x_0$ in (1) gives:

$$y^{(1)}(x_0) + p(x_0)y(x_0) = r(x_0)$$

$\Rightarrow$ But $y(x_0) = y_0$, thus;

$$\begin{aligned} y^{(1)}(x_0) + p(x_0)y_0 &= r(x_0) \\ \Rightarrow y^{(1)}(x_0) &= r(x_0) - p(x_0)y_0 \end{aligned}$$

$\Rightarrow$ To determine $y^{(2)}(x_0)$, we differentiate (1) once with respect to $x$ to obtain:

$$y^{(2)}(x) + p^{(1)}(x)y(x) + p(x)y^{(1)}(x) = r^{(1)}(x)$$

$\Rightarrow$ Where $p^{(1)}(x) = p'(x)$ and $r^{(1)}(x) = r'(x)$. Then aftre setting $x = x_0$ and using $y^{(1)}(x_0) = r(x_0) - p(x_0)y_0$, we get;

$$y^{(2)}(x_0) = r^{(1)}(x_0) - p^{(1)}(x_0)y_0 - p(x_0)[r(x_0) - p(x_0)y_0]$$

10) Higher order derivatives $y^{(n)}(x_0)$ can be computed in similar fashion by repeated differentaition of the original differential equation coupled with the use of lower order derivatives that have already been determined.

11) Once the values of $y^{(k)}(x_0)$ have been found for $k = 1, 2, 3, \cdots, N$ for some given integer $N$, substitution into series (3) provides the required approximation to the power series solution of the initial value problem for the differential equation up to terms of the order $(x - x_0)^N$.

12) This method generates the Taylor Series expansion of $y(x)$ about the point $x_0$ when $x_0 \neq 0$, and its Maclaurin Series expansion when $x_0 = 0$, though these series are often simply called **Power Series** about $x_0 \neq 0$ and $x_0 = 0$, respectively.

## EXAMPLES

1. (a) Determine whether $x = 0$ is an ordinary point of the following differential equations:
   (i) $y'' - xy' + 2y = 0$
   (ii) $y'' + y = 0$
   (iii) $2x^2y'' + 7x(x + 1)y' - 3y = 0$
   (iv) $x^2y'' + 2y' + xy = 0$

   (b) Determine whether $x = 0$ or $x = 1$ is an ordinary point of the differential equation:

$$(1 - x^2)y'' - 2xy' + n(n + 1)y = 0$$

**SOLUTION**

(a) (i) $y'' - xy' + 2y = 0$

$\Rightarrow$ Here $p(x) = -x$ and $q(x) = 2$ are both polynomials, hence they are analytic everywhere.
$\Rightarrow$ Therefore, every value of $x$, in particular $x = 0$, is an ordinary point.

(ii) $y'' + y = 0$
$\Rightarrow$ Here $p(x) = 0$ and $q(x) = 1$ are both constants, hence they are analytic everywhere.
$\Rightarrow$ Therefore, every value of $x$, in particular $x = 0$, is an ordinary point.

(iii) $2x^2y'' + 7x(x+1)y' - 3y = 0$

$\Rightarrow$ Dividing by $2x^2$, we get $p(x) = \frac{7(x+1)}{2x}$ and $q(x) = -\frac{3}{2x^2}$.

$\Rightarrow$ As neither function is analytic at $x = 0$ (both denominators are zero there), $x = 0$ is not an ordinary point but rather, a singular point.

(iv) $x^2y'' + 2y' + xy = 0$

$\Rightarrow$ Here $p(x) = \frac{2}{x^2}$ and $q(x) = \frac{1}{x}$. Neither of these functions is analytic at $x = 0$, so $x = 0$ is not an ordinary point, but a singular point.

(b) $(1 - x^2)y'' - 2xy' + n(n+1)y = 0$

$\Rightarrow$ We first transform the differential equation into the form of equation (2) by dividing by $(1 - x^2)$, then we get:

$$p(x) = -\frac{2x}{1 - x^2} \quad \text{and} \quad q(x) = \frac{n(n+1)}{1 - x^2}$$

$\Rightarrow$ Both of these functions have taylor series expansions around $x = 0$, so both are analytic there and $x = 0$ is an ordinary point.
$\Rightarrow$ In contrast, the denominators of both functions are zero at $x = 1$, so neither function is defined there and therefore is analytic there. Consequently, $x = 1$ is a singular point.

2. Find the first five terms in series solution of:

$$y' + (1 + x^2)y = \sin x, \quad \text{with} \quad y(0) = a$$

**SOLUTION**

$\Rightarrow$ As the initial condition is specified at $x = 0$, the power series solution is an expansion about the origin and so is a **Maclaurin Series**.

$\Rightarrow$ The functions $(1 + x^2)$ and $\sin x$ are analytic for all $x$, so the series expansion can be found about the origin.

$\Rightarrow$ Setting $x = 0$ in the equation and substituting for the initial conditions shows that:

$$y'(0) = y^{(1)}(0) = -a$$

⇒ Differentiating the differential equation gives:

$$y^{(2)} + 2xy + (1 + x^2)y^{(1)} = \cos x$$

⇒ Where $y^{(2)} = y''$, so setting $x = 0$, we get:

$$y^{(2)}(0) + y^{(1)}(0) = 1$$

⇒ But $y^{(1)}(0) = -a$ and so $y^{(2)}(0) = 1 + a$.

⇒ Repeating this process to find higher order derivatives leads to the results:

$$y^{(3)}(0) = -(1 + 3a) \quad \text{and} \quad y^{(4)}(0) = 9a$$

⇒ Substituting these results into series (3) shows that, to terms of order $x^4$, the required solution takes the form:

$$y(x) = a - ax + \frac{(1 + a)}{2!}x^2 - \frac{(1 + 3a)}{3!}x^3 + \frac{9a}{4!}x^4 + \cdots$$

3. Find the first five terms in series solution of:

$$y' + 4xy = e^{x-1}, \quad \text{with} \quad y(1) = 1$$

**SOLUTION**

⇒ In this case the functions $4x$ and $3e^{x-1}$ are analytic for all, but as the expansion is about $x = 1$, the power series solution that is obtained will be a Taylor series expansion about the point $x = 1$.

⇒ Setting $x = 1$ in the differential equation and using the initial condition $y(1) = 1$, we get:

$$y^{(1)}(1) = -1$$

⇒ Differentiating the differential equation gives:

$$y^{(2)} + 4y + 4xy^{(1)} = 3e^{x-1}$$

⇒ Setting $x = 1$ and using the result $y^{(1)}(1) = -1$, we get:

$$y^{(2)}(1) = 3$$

⇒ Repeating this process leads to the results that:

$$y^{(3)}(1) = -1 \quad \text{and} \quad y^{(4)}(1) = -29$$

⇒ Substituting into (3) shows that the Taylor series expansion of the solution up to terms of order $(x - x_0)^4$ is:

$$y(x) = 1 - (x - 1) + \frac{3}{2}(x - 1)^2 - \frac{1}{6}(x - 1)^3 - \frac{29}{24}(x - 1)^4 + \cdots$$

4. Find the terms up to $x^5$ in the series solution of:

$$y'' + xy' + (1 - x^2)y = x, \quad \text{with} \;\; y(0) = a \;\; \text{and} \; y'(0) = b$$

**SOLUTION**

$\Rightarrow$ The coefficients $x$ and $(1 - x^2)$ and the nonhomogeneous term $x$ are analytic for all $x$, so as the initial data is given at $x = 0$, Maclaurin Series solution can be found.

$\Rightarrow$ Setting $x = 0$ in the equation and using the initial conditions $y(0) = a$ and $y^{(1)} = b$ gives:

$$y^{(2)}(0) = -a$$

$\Rightarrow$ Differentiating the diferential equation, we get:

$$y^{(3)} + y^{(1)} + xy^{(2)} - 2xy + (1 - x^2)y^{(1)} = 1$$

$\Rightarrow$ Setting $x = 0$ and using the results $y^{(2)}(0) = -a$ and $y^{(1)}(0) = b$, we get:

$$y^{(3)}(0) = 1 - 2b$$

$\Rightarrow$ a repetition of this process leads to the results:

$$y^{(4)}(0) = 5a \;\; \text{and} \;\; y^{(5)}(0) = 14b - 4$$

$\Rightarrow$ Substituting these results into (3) shows that to terms of order $x^5$, the Maclaurin series expansion of the solution is:

$$y(x) = a + bx - \frac{a}{2}x^2 + \frac{(1 - 2b)}{6}x^3 = \frac{5a}{24}x^4 + \frac{(7a - 2)}{60}x^5 + \cdots$$